

Institute of Actuaries of India

Subject CB2-Business Economics

September 2021 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

<u>Solution 1:</u>	D	[1.5]
<u>Solution 2:</u>	D	[1.5]
<u>Solution 3:</u>	D	[1.5]
<u>Solution 4:</u>	D	[1.5]
<u>Solution 5:</u>	A	[1.5]
<u>Solution 6:</u>	C	[1.5]
<u>Solution 7:</u>	A	[1.5]
<u>Solution 8:</u>	B	[1.5]
<u>Solution 9:</u>	D	[1.5]
<u>Solution 10:</u>	A	[1.5]
<u>Solution 11:</u>	C	[1.5]
<u>Solution 12:</u>	B	[1.5]
<u>Solution 13:</u>	B	[1.5]
<u>Solution 14:</u>	D	[1.5]
<u>Solution 15:</u>	D	[1.5]
<u>Solution 16:</u>	C	[1.5]
<u>Solution 17:</u>	D	[1.5]
<u>Solution 18:</u>	B	[1.5]
<u>Solution 19:</u>	A	[1.5]
<u>Solution 20:</u>	B	[1.5]
<u>Solution 21:</u>	A	[1.5]
<u>Solution 22:</u>	B	[1.5]
<u>Solution 23:</u>	B	[1.5]
<u>Solution 24:</u>	B	[1.5]
<u>Solution 25:</u>	A	[1.5]
<u>Solution 26:</u>	D	[1.5]

Solution 27: Every country has a central bank. The central bank is usually a government financial institution with a number of important responsibilities:

- Banker to the government. The central bank acts as a banker to the government in ways that are similar to how commercial banks act as bankers to their customers. It holds the government's cash (as deposits), it receives payments for the government and makes payments for the government (the government writes cheques (checks) paid from an account in the central bank). It also manages the government's borrowing by selling bonds to commercial banks⁶ and the public, and acts as an adviser to the government on financial and banking matters.
- Banker to commercial banks. The central bank also acts as a banker to commercial banks by holding deposits for them and can also make loans to them in times of need. (The central bank does not act as a banker to consumers and firms.)
- Regulator of commercial banks. The central bank regulates and supervises commercial banks, making sure they operate with appropriate levels of cash and according to rules that ensure the safety of the financial system. This is a very important function, because the funds that commercial banks use to make loans are the savings and other deposits that consumers and firms deposit with commercial banks.
- Conduct monetary policy. The central bank is responsible for monetary policy, based on changes in the supply of money or the rate of interest. The central bank is also usually responsible for the determination of exchange rates (the price of the

domestic currency in terms of foreign currencies) because of the close relationship between interest rates and exchange rates.

(1 marks for each explanation)

[4]

Solution 28:

- i) Deregulation- Deregulation involves the elimination or reduction of government regulation of private sector activities, and is based on the argument that government regulation stifles competition and increases inefficiency. There are two main types of regulation (and deregulation): economic and social. 'Economic regulation' involves government control of prices, output, and other activities of firms, offering them protection against competition. In the last two to three decades, many countries have moved toward removal of government regulations, and hence economic deregulation. A main form of deregulation has been to allow new, private firms to enter into monopolistic or oligopolistic industries, thus forcing existing firms to face competition. The objective has been to increase efficiency, lower costs and improve quality. Industries affected include transport, airlines, television broadcasting, telecommunications, natural gas, electricity, financial services and others. 'Social regulation' involves protecting consumers against undesirable effects of private sector activities (many of these involve negative externalities) in numerous areas, including food, pharmaceutical and other product safety, worker protection against injuries, and pollution control. In contrast to economic regulation, social regulation is being strengthened in many countries in the interests of public safety. Some economists, however, argue that social regulation is excessive, giving rise to costly and inefficient bureaucratic procedures, paperwork and unnecessary government interference, and should therefore be reduced.
- ii) Privatisation- Privatisation, involving a transfer of ownership of a firm from the public to the private sector, can increase efficiency due to improved management and operation of the privatised firm. This is based on the argument that government enterprises are often inefficient as they have bureaucratic procedures, high administrative costs and unproductive workers, because they do not face incentives to lower costs and maximise profits. The private sector may therefore be more efficient than the public sector.
- iii) Trade liberalization- International trade between countries has become freer in recent decades due to reductions in trade barriers. According to economic theory, free or freer trade increases competition between firms both domestically and globally, resulting in greater efficiency in production and an improved allocation of resources.
- iv) Restricting monopoly power- Increased competition can result from restricting monopoly power of firms by enforcing anti-monopoly legislation, by breaking up large firms that have been found to engage in monopolistic practices into smaller units that will behave more competitively, and by preventing mergers between firms that might result in too much monopoly power. Greater scope for the forces of supply and demand may result in increased efficiency, lower costs and improved quality.

[2]

[2]

[2]

[2]

(2 marks for each subsections)

[8]

Solution 29: The benefits of trade are:

Increases in domestic production and consumption as a result of specialisation

Many of the benefits of trade arise from specialisation. Specialisation occurs when an individual, firm or country concentrates production on one or a few goods and services. A country that does not trade must itself produce all the goods and services consumed, and therefore cannot specialise. However, if it uses its resources to specialise in the production of those goods and services it can produce more efficiently (with lower costs of production), it can produce more of these, and trade some of them for other goods produced more efficiently in other countries. This way it is able to produce a greater quantity of output because it does not 'waste' its scarce resources on producing goods and services at a relatively high cost. It can also increase its consumption of goods and services, because by exporting part of its larger domestic output in exchange for other output produced more cheaply elsewhere, it can acquire a larger overall quantity of goods and services.

Economies of scale in production

The ability of firms to decrease average costs of production (cost per unit of output) by becoming larger and increasing the quantity of output produced. When a firm lowers its average costs, it becomes more efficient, and can sell its output at a lower price. In the absence of trade, the amount of output any firm can produce is limited by the size of the domestic market. If the domestic market is small (if the country is small), the firm is unable to grow and take advantage of economies of scale. The possibility of trade and exports to other countries involves an expansion in the size of the market, allowing firms to produce more output, achieve economies of scale and enjoy the benefits of lower costs, which include lower prices and therefore greater export competitiveness, or the ability to compete better in foreign markets.

Greater choice for consumers

The goods and services each country can produce differ widely with respect to their variety and their quality. By trading with each other, countries can import a larger variety of goods and services, possibly of higher quality, than the ones they can produce themselves. This increases choice for consumers.

Increased competition and greater efficiency in production

When countries trade with each other, domestic firms become exposed to competition from products produced by firms in other countries. They are therefore forced to become more efficient; in other words, they must try to produce at the lowest possible cost. If they do not become more efficient, they will have to sell their output at higher prices to cover their higher costs; consumers will prefer the lower-priced imported products, and higher cost firms may go out of business. Therefore, increased competition leads to greater efficiency.

Lower prices for consumers

Increased competition and efficiency among firms leads to lower prices for consumers. In addition, as imports consist of goods that are produced more

efficiently in other countries, this is an additional factor leading to lower prices for consumers.

Acquiring needed resources

Countries may need for their domestic production a number of natural resources or capital goods that are not available domestically. For example, oil is a resource that virtually all countries depend on, yet most are forced to rely on imported oil because they do not produce it themselves. The same may apply to a variety of other resources such as timber, minerals and semi-finished products used as inputs, as well as capital goods (machinery and equipment) used in production. Trade allows countries to import inputs they need for domestic production.

Free trade and a more efficient allocation of resources and is a source of foreign exchange

If trade is free, meaning that there is no government intervention imposing restrictions on trade, it can lead to a more efficient allocation of resources.

When countries sell goods and services to other countries, they acquire foreign exchange (or foreign currencies), which allows them to make payments to other countries for the goods and services they import, or make other payments abroad.

Trade makes countries interdependent, reducing the possibility of hostilities and violence Strong international trade links between countries can form the basis for economic relationships that reduce the possibility of war or other hostilities. For example, one of the reasons behind the establishment of the European Economic Community in 1957 (the EEC, the precursor of the European Union) was to eliminate the possibility of future wars between France and Germany. The strong economic interdependence created by trade (and other) links between these countries makes the possibility of war between them inconceivable today.

Trade makes possible the flow of new ideas and technology and is an 'engine for growth'

As goods and services flow from one country to another, they enable new ideas and new technologies and skills to be transferred from one country to another. Increased specialisation, economies of scale, greater efficiencies in production, acquisition of needed resources, increased competition, technological advances and expanding markets, all made possible by international trade, contribute to increases in domestic output, and therefore to greater economic growth. Because of its major potential contributions to economic growth, international trade has been termed an 'engine for growth'.

(Explaining any 5 relevant points)

[10]

Solution 30:

- i) Aggregate demand is the total amount of real output (real GDP) that consumers, firms, the government and foreigners want to buy at each possible price level, over a particular time period. The aggregate demand (AD) curve shows the relationship between the total amount of real output demanded by the four components and the economy's price level over a particular time period. It is downward-sloping, indicating a negative relationship between the price level and aggregate output demanded.

[2]

- ii) The reasons behind the downward slope of the aggregate demand are very different from demand in a single market in microeconomics. They include the following:
- The wealth effect. Changes in the price level affect the real value of people's wealth. (Wealth is not the same as income; wealth is the value of assets that people own, including their houses, stocks and bonds, their jewellery, works of art, and so on.) If the price level increases, the real value of wealth falls. People feel worse off and cut back on their spending on goods and services. Therefore, as the price level increases, less output is demanded, leading to an upward movement along the AD curve. If there is a fall in the price level, the real value of wealth increases, people feel better off and increase their spending, thus more output is demanded, causing a downward movement along the AD curve.
 - The interest rate effect. Changes in the price level affect rates of interest, which in turn affect aggregate demand. If there is an increase in the price level, consumers and firms need more money to carry out their purchases and transactions. This leads to an increase in the demand for money, which in turn leads to an increase in rates of interest. As interest rates rise, the cost of borrowing increases leading to a decrease in consumer purchases financed by borrowing, as well as investment spending by firms that must borrow to finance their expenditures. Therefore, increases in the price level lead to a fall in quantity of output demanded, or an upward movement along the AD curve. A fall in the price level leads to a rise in quantity of output demanded, or a downward movement along the AD curve.
 - The international trade effect. If the domestic price level increases while price levels in other countries remain the same, exports become more expensive to foreign buyers who will now demand a smaller quantity of these. At the same time, goods produced in other countries become relatively cheaper, so domestic buyers increase their purchases (imports) from foreign countries. Therefore, a rising price level produces a fall in exports and a rise in imports so that net exports, $X-M$, fall. Falling net exports represent a fall in quantity of output demanded or an upward movement along the AD curve. A fall in the domestic price level relative to other countries leads to a larger amount of exports demanded and lower amount of imports demanded, so that net exports, $X-M$, rise. Therefore, there is a downward movement along the AD curve.

(3 points * 1 mark)

[3]

[5]

Solution 31:

- i)
Industrial clusters are group of interconnected companies that are located in same geographical region.
- ii)
The government policies for development of industrial clusters are as follows:
1. Lowering business rates to companies operating within industrial clusters
 2. Simplified planning regulations for the firms
 3. Government funding for research and development for the firms

[1]

4. To increase productivity and innovation by encouraging spread of ideas between firms by not offering patents.

(Half mark for each point, 4*0.5) [2]

iii)

Main features of Austrian School are as follows:

- The process of competition-
 - a) Preferences of consumers are complex and are not rational in conventional sense.
 - b) Different preference of consumers can not be known by government making central planning of economy impossible.
 - Government Intervention-
 - a) Austrian school does not support government regulation that directs economic activity. They support role of the market rather than the government regulation.
 - b) System of regulation evolve within market itself
- Central bank's loose monetary policy would lead to distortion of economy and inflation.

(Any four points – each point carries 1 mark) [4]

[7]

Solution 32:

- i) The market structure is "Perfect Competition".

[1]

- ii) Description of Perfect Competition-

It is a market structure in which :

- a) there are many firms
- b) there is freedom of entry to industry
- c) all firms produce an identical product
- d) all firms are price takers

(Each point carries half mark) [2]

iii)

Consumers prefer this type of market structure because-

- In perfect market structure economic efficiency is achieved when-
 - a) Output is product at minimal cost (productive efficiency)
 - b) Consumers get maximum benefit from their income (allocative efficiency)
- Productive efficiency-It is achieved if firms produce maximum output for a given amount of input
- Allocative efficiency- It is achieved when a firm produces at socially optimal level of output. i.e. resources are allocated in such a way to maximise welfare.
- The combination of goods produced and sold gives the maximum satisfaction for each consumer at their current levels of income
- Consumers benefit from allocative efficiency and productive efficiency.
- In a long run, the firm earns only normal profits so prices are reasonable and consumers are not exploited

(Each point carries 1 mark) [6]

[9]

Solution 33:

i)

Definition- Price ceiling-

Price ceiling operates when Government passes a law making it illegal to charge more than a certain amount for a good. [1]

Advantages:

1. The people can able to afford the drugs who could not do so under the free market condition.
2. Drugs are essential during pandemics, otherwise they would be very expensive and unaffordable for the poor.

Disadvantages:

1. Some suppliers may be tempted to offer goods for sale illegally at a price, above price ceiling i.e. shadow market may develop.
2. This may encourage criminality

There will be cost of enforcement involved in trying to prevent this.

(2 advantages and any 2 disadvantages, Total 4)

[5]

ii)

As more units of labour are employed, the marginal physical product of labour may increase at first. But tend eventually fall due to the law of diminishing marginal productivity.

(Each point carries 1 mark)

[2]

iii)

Relationship between marginal physical product of labour curve and the marginal cost curve:

1. Assuming wage rate is constant, when the marginal physical product of labour curve is increasing, the marginal cost curve is decreasing
2. When the marginal physical product of labour curve is decreasing, the marginal cost curve is increasing.

(Each point carries 1 mark)

[2]

[9]

Solution 34:

It is likely to be relatively high for the car insurance industry because:

- the industry is oligopolistic
- car insurance is a relatively high expenditure item
- car insurance needs to be renewed annually and so there is a continuous flow of buyers for the product
- consumers are increasingly looking to switch between providers in order to obtain a better deal rather than sticking with their current provider
- consumers need to be made aware of the products and product features that are currently available, as car insurance is a relatively infrequent purchase (usually only once a year) and product specifications change frequently
- car insurance providers attempt to differentiate their product from those of their competitors, eg by offering better service standards (such as roadside assistance and courtesy cars)

as car insurance is intangible, potential consumers cannot “see” it in the same way they can view a car for example, and so advertising is necessary to raise the profile of a particular car insurance product.

(Each point carries 1 mark)

[Max 5]

Solution 35:

“Standard of living” is a broader term than GDP. While GDP focuses on production that is bought and sold in markets, standard of living includes all elements that affect people’s well-being, whether they are bought and sold in the market or not. [1]

- While GDP includes spending on recreation and travel, it does not cover leisure time. However, there is a substantial difference between an economy that is large because people work long hours, and an economy that is just as large because people are more productive with their time so they do not have to work as many hours.
- While GDP includes what a country spends on environmental protection, healthcare, and education, it does not include actual levels of environmental cleanliness, health, and learning. GDP includes the cost of buying pollution control equipment, but it does not address whether the air and water are actually cleaner or dirtier.
- GDP includes spending on medical care, but does not address whether life expectancy or infant mortality have risen or fallen.
- Similarly, it counts spending on education, but does not address directly how much of the population can read, write, or do basic mathematics.
- GDP includes production that is exchanged in the market, but it does not cover production that is not exchanged in the market. For example, hiring someone to mow your lawn or clean your house is part of GDP, but doing these tasks yourself is not part of GDP.
- GDP has nothing to say about the level of inequality in society. GDP per capita is only an average. When GDP per capita rises by 5%, it could mean that GDP for everyone in the society has risen by 5%, or that GDP of some groups has risen by more while that of others has risen by less—or even declined.
- GDP has nothing much to say about what technology and products are available. The standard of living in, for example, 1950 or 1900 was not affected only by how much money people had—it was also affected by what they could buy. No matter how much money you had in 1950, you could not buy an iPhone or a personal computer.
- In certain cases, it is not clear that a rise in GDP is even a good thing. If a city is wrecked by a hurricane, and then experiences a surge of rebuilding construction activity, it would be peculiar to claim that the hurricane was therefore economically beneficial. If people are led by a rising fear of crime, to pay for installing bars and burglar alarms on all their windows, it is hard to believe that this increase in GDP has made them better off.

A high level of GDP should not be the only goal of macroeconomic policy, or government policy more broadly. Even though GDP does not measure the broader standard of living with any precision, it does measure production well and it does indicate when a country is materially better or worse off in terms of jobs and incomes. In most countries, a significantly higher GDP per capita occurs hand in hand with other improvements in everyday life along many dimensions, like education, health, and environmental protection.

(Any of the 6 bullet points = $6 \times 0.5 = 3$ marks)

[4]
